**Date:**15/10/2026

# CO1-Assignment Tool 3
# Coding Challenge

**Course code:** DSA0308
**Course Name:** Natural Language Processing

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### 1. Intelligent email and password validator using Regular Expressions

```
import re
email = input("Enter Email Address: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")
email_pattern = r"^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$"
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$%&!]).{8,}$"
mobile_pattern = r"^[6-9]\d{9}$"
if re.match(email_pattern, email):
  print("Valid Email")
else:
  print("Invalid Email")
if re.match(password_pattern, password):
  print("Strong Password")
else:
  print("Weak Password")
if re.match(mobile_pattern, mobile):
  print("Valid Mobile Number")
else:
  print("Invalid Mobile Number")
```

**Output:**

```
= RESTART: C:/Users/RAIHAN/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y
Enter Email Address: raihan0321@gmail.com
Enter Password: 12345
Enter Mobile Number: 8939643229
Valid Email
Weak Password
Valid Mobile Number
```

### 2. Design a Finite State Automata (FSA) simulator

```
states = input("Enter states: ").split()
alphabet = input("Enter input alphabet: ").split()
transition = {}
print("Enter transition table:")
for state in states:
  transition[state] = {}
  for symbol in alphabet:
    next_state = input(f"Delta({state}, {symbol}) = ")
    transition[state][symbol] = next_state
initial = input("Enter initial state: ")
final_states = input("Enter final states: ").split()
n = int(input("Enter number of input strings: "))
for i in range(n):
  string = input("Enter input string: ")
  current = initial
  path = [current]
  valid = True
  for symbol in string:
    if symbol not in alphabet:
```

```
      valid = False
      break
    current = transition[current][symbol]
    path.append(current)
  print("Transition Path:")
  print(" -> ".join(path))
  if valid and current in final_states:
    print("Accepted")
  else:
    print("Rejected")
```

**Output:**

```
= RESTART: C:/Users/RAIHAN/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y
Enter states: q0 q1 q2
Enter input alphabet: a b
Enter transition table:
Delta(q0, a) = q1
Delta(q0, b) = q0
Delta(q1, a) = q1
Delta(q1, b) = q2
Delta(q2, a) = q1
Delta(q2, b) = q0
Enter initial state: q0
Enter final states: q2
Enter number of input strings: 2
Enter input string: abab
Transition Path:
q0 -> q1 -> q2 -> q1 -> q2
Accepted
Enter input string:
```

**3. Smart Pattern Matching Engine using Regular Expressions**

```
import re
text = """Meeting on 12/09/2026
Call 8939643229
#NLP
@OpenAI
natural language processing"""
while True:
  print("\nUser Menu")
  print("1. Search Word")
  print("2. Search Date")
  print("3. Search Phone Number")
  print("4. Search Hashtag")
  print("5. Search Mention")
  print("6. Search Prefix")
  print("7. Search Suffix")
  print("8. Exit")
  choice = input("Enter your choice: ")
  if choice == "1":
    word = input("Enter word to search: ")
    pattern = r"\b" + re.escape(word) + r"\b"
  elif choice == "2":
    pattern = r"\b\d{2}/\d{2}/\d{4}\b"
  elif choice == "3":
    pattern = r"\b[6-9]\d{9}\b"
  elif choice == "4":
    pattern = r"#[A-Za-z0-9_]+"
  elif choice == "5":
    pattern = r"@[A-Za-z0-9_]+"
  elif choice == "6":
```

```
    prefix = input("Enter prefix: ")
    pattern = r"\b" + re.escape(prefix) + r"\w*"
  elif choice == "7":
    suffix = input("Enter suffix: ")
    pattern = r"\w*" + re.escape(suffix) + r"\b"
  elif choice == "8":
    print("Exiting...")
    break
  else:
    print("Invalid choice")
    continue
  matches = re.findall(pattern, text, re.IGNORECASE)
  if matches:
    print("Matching Patterns:")
    for match in matches:
      print(match)
  else:
    print("No matching patterns found")
```

**Output:**

```
= RESTART: C:/Users/RAIHAN/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y

User Menu
1. Search Word
2. Search Date
3. Search Phone Number
4. Search Hashtag
5. Search Mention
6. Search Prefix
7. Search Suffix
8. Exit
Enter your choice: 1
Enter word to search: Clean
No matching patterns found

User Menu
1. Search Word
2. Search Date
3. Search Phone Number
4. Search Hashtag
5. Search Mention
6. Search Prefix
7. Search Suffix
8. Exit
Enter your choice: 2
Matching Patterns:
12/09/2026
```